**Supplementary Material**

# Soil Hydraulic Memory and the Recoverability of Evaporation Savings and the Closure of the Water Balance in a Simulation Study with Daisy

Derivations of the equations, the mechanisms behind the results, the data, and the origin of every table and figure

**Naziru Halilu**[1,2,*]

[1] Agricultural University of Athens, Athens, Greece

[2] Public University of Navarre, Pamplona, Spain

ORCID: 0009-0006-2324-9599

[*] Correspondence: halilunaziru73@gmail.com

## Contents

**S1 Derivations of the equations** **3**

S1.1 The retention and conductivity functions . . . . . 3

S1.2 The water balance and the storage release . . . . . 3

S1.3 The memory fate as a pooled within-group slope . . . . . 3

S1.4 Loss fates and the recoverable share . . . . . 4

S1.5 The rate-competition law . . . . . 4

S1.6 The invariant search by singular-value decomposition . . . . . 4

S1.7 The closure residual and the unattributed term . . . . . 5

S1.8 Log-ratio coordinates and the sparse law . . . . . 5

S1.9 Irrigation compensation . . . . . 5

**S2 Why the results happen** **5**

**S3 The data** **6**

**S4 Where every table and figure comes from** **31**

## S1. Derivations of the equations

### S1.1 The retention and conductivity functions

With the capillary law $r \propto 1/h$, Mualem (1976) expressed the relative conductivity of a bundle of connected capillaries as

$$K_r(S_e) = S_e^{\lambda} \left[ \int_0^{S_e} h(x)^{-1} \mathrm{d}x \Big/ \int_0^{1} h(x)^{-1} \mathrm{d}x \right]^2 . \tag{S1}$$

Inserting the inverse of the van Genuchten curve, $h(x) = \alpha^{-1}(x^{-1/m} - 1)^{1/n}$, and substituting $y = x^{1/m}$ gives $\int_0^{S_e} h^{-1}\mathrm{d}x = \alpha m \int_0^{S_e^{1/m}} y^{m-1+1/n}(1-y)^{-1/n}\mathrm{d}y$. With $m = 1 - 1/n$ the power of $y$ vanishes and the integral is $\alpha[1 - (1 - S_e^{1/m})^m]$; dividing by its value at $S_e = 1$ ($\alpha$) and squaring yields the Mualem conductivity of the paper, $K = K_s S_e^{\lambda}[1 - (1 - S_e^{1/m})^m]^2$. For large $\alpha h$, $S_e \simeq (\alpha h)^{-(n-1)}$ and $K_r \simeq m^2(\alpha h)^{-[2n+\lambda(n-1)]}$: the conductivity of a soil with large $n$ falls faster with suction, which is the property that orders hydraulic memory.

### S1.2 The water balance and the storage release

Over a period, mass conservation of the soil column with surface runoff $Q_r$ is $P_r + I = ET + D + \Delta S + Q_r$. Rearranging,

$$\bar{R} \equiv ET + D - P_r - I = -(\Delta S + Q_r), \tag{S2}$$

so the storage release is positive when the column loses stored water (or sheds no runoff) and supplies the fluxes. In the initial-moisture experiment, a wetter initial state raises the stored water that is released over the run, which is the variation used to measure the memory fates.

### S1.3 The memory fate as a pooled within-group slope

Within a group of runs with the same rainfall scale $P$, write each flux as $x_c = a_P + \phi_x \bar{R}_c + e_c$. Minimising $\sum_P \sum_{c \in P} e_c^2$ over the group intercepts $a_P$ gives $a_P = \bar{x}_P - \phi_x \overline{\bar{R}}_P$, and then over $\phi_x$

$$\phi_x = \frac{\sum_P \sum_{c \in P} (\bar{R}_c - \overline{\bar{R}}_P)(x_c - \bar{x}_P)}{\sum_P \sum_{c \in P} (\bar{R}_c - \overline{\bar{R}}_P)^2}. \tag{S3}$$

Because the four fluxes add to $ET + D = \bar{R} + P_r + I$ and $P_r + I$ is constant within a group, the deviations of the four fluxes add to the deviation of $\bar{R}$, and the four slopes add to one: $\phi_{E_s} + \phi_T + \phi_D + \phi_{\text{oth}} = 1$. Each $\phi_x$ is therefore the share of a released millimetre that leaves by flux $x$.

### S1.4 Loss fates and the recoverable share

Between two levels of the evaporation factor the change of the non-transpirational evapotranspiration $\delta ET_{\text{nt}}$ must be balanced by the other fluxes. With storage changes negligible over 20 years, $\delta ET_{\text{nt}} + \delta T + \delta D - \delta I \simeq 0$, hence

$$\psi_T + \psi_D + \psi_I \simeq 1, \qquad \psi_x = \mp \frac{\delta x}{\delta ET_{\text{nt}}}, \tag{S4}$$

the signs chosen so that each $\psi$ is positive when the flux compensates the evaporation change. Of a millimetre of evaporation saved, $\psi_D$ leaves as drainage and is not recovered by the crop or the irrigation account; the recoverable share is $\rho_R = 1 - \psi_D$.

### S1.5 The rate-competition law

Let stored water $S$ above field capacity leave by two parallel first-order sinks, drainage with rate constant $k_D \propto K_s/L$ and withdrawal by evaporation and uptake with $k_W \propto q^*/L$, where $L$ is a common length. Then $\mathrm{d}S/\mathrm{d}t = -(k_D + k_W)S$ and the drainage leaving over the release is

$$\int_0^\infty k_D S \,\mathrm{d}t = \frac{k_D}{k_D + k_W} S_0 \quad \Longrightarrow \quad \phi_D = \frac{K_s}{K_s + q^*}. \tag{S5}$$

The law is bounded in $[0, 1]$, tends to 0 for an impermeable soil and to 1 for a free-draining one, and is the only one-parameter form with these limits that is linear in $K_s$ at small $K_s$. By the Buckingham theorem, $\phi_D = F(K_s/q^*, n, \alpha h^*)$; the law keeps the first group only.

### S1.6 The invariant search by singular-value decomposition

Arrange the 20-year mean fluxes of all runs as the rows of a matrix $\mathbf{F}$ (columns $P_r, I, ET, D$, centred). A linear invariant is a vector $\mathbf{c}$ with $\mathbf{Fc} \approx \mathbf{0}$. The best such vector in the least-squares sense, with $\|\mathbf{c}\| = 1$, minimises $\|\mathbf{Fc}\|^2 = \mathbf{c}^\top \mathbf{F}^\top \mathbf{F}\mathbf{c}$, which is the eigenvector of $\mathbf{F}^\top\mathbf{F}$ with the smallest eigenvalue, that is the right singular vector of $\mathbf{F}$ with the smallest singular value. Rescaled so that the *ET* coefficient is $-1$, it is the discovered balance; the permutation null shuffles each col-

umn independently and so destroys any true invariant, which gives the distribution of the smallest singular value under no structure.

### S1.7 The closure residual and the unattributed term

With $W = P_r + I$, the residual $\varepsilon = (W - ET - D)/W \simeq (\Delta S + Q_r)/W$ is the share of the input not leaving by evapotranspiration or drainage. The unattributed evapotranspiration is $R = ET - T - E_s - E_c$; in the full outputs it equals the evaporation of ponded water, litter and snow (data folder, closure within 1.1 mm).

### S1.8 Log-ratio coordinates and the sparse law

Fluxes that share a total are compositional; the centred log-ratio $\mathrm{clr}(\mathbf{x})_i = \ln x_i - \frac{1}{5}\sum_j \ln x_j$ removes the constraint. The sparse laws regress $\ln y$ on a small set of log or dimensionless candidates chosen by forward selection; out-of-sample skill is $R^2_{\mathrm{cv}} = 1 - \sum(\ln y_i - \widehat{\ln y_i}^{(-g)})^2 / \sum(\ln y_i - \overline{\ln y})^2$, with each prediction made by a model fitted without the group $g$ (experiment or soil) of run $i$.

### S1.9 Irrigation compensation

With the same balance, a change of rainfall $\Delta P_r$ at fixed trigger is offset partly by irrigation: $\kappa = -\Delta I / \Delta P_r$. When drainage and evapotranspiration do not change, $\kappa = 1$; when part of the extra rain drains, $\kappa < 1$.

## S2. Why the results happen

*Why hydraulic memory differs between soils.* Water released from storage leaves by the faster of two routes. In coarse soils (large $n$, high $K_s$) the conductivity at field capacity is high, so released water drains before roots and the surface can use it; in fine soils the conductivity is low and the same water stays near the roots and the surface, where it is transpired or evaporated.

*Why the memory transfers to other experiments.* The same competition decides where any extra water goes, whether it comes from initial storage, from a lower evaporation factor or from a mulch; the share is a property of the soil's conductivity, not of the source of the water.

*Why warming raises and wetter climates lower the recoverable share.* Warming raises the evaporative demand, so the withdrawal side of the competition is faster and less of a saved millimetre drains. A wetter climate keeps the profile closer to field capacity, where drainage is fast, so more of a saving drains.

*Why nitrogen has no effect.* The nitrogen rate changes the crop's growth but not the hydraulic properties, and in these runs it changes the water fluxes too little to move the shares; it therefore serves as a negative control.

*Why an evaporative term is unattributed.* The reported components cover transpiration, soil evaporation and canopy evaporation; evaporation of ponded water, litter and snow is part of evapotranspiration but not of those components, so it appears as a residual in the run table.

*Why evapotranspiration follows a compact law and drainage does not.* Evapotranspiration is limited by energy and by water and is smooth in its inputs over 20 years. Drainage is a threshold process (water drains only above field capacity) whose annual total depends on the timing of rain and irrigation events, which the long-term means do not contain.

*Why judging a saving against soil evaporation alone misleads.* A mulch changes soil evaporation and also the unattributed evaporation; if only the reported soil evaporation is used as the denominator, the drainage share of the change is computed against the wrong total and can change sign.

## S3. The data

Table S1 lists the 17 experiments of the data folder and Fig. S1 shows their outputs. The tables that follow are the result files of the analyses of the paper, printed in full or up to 220 rows; each is also in the workbook `Data/Manuscript_Data/SWAFL_Manuscripts_Data.xlsx`, and every run is in `Data/Comprehensive_Summary`.

**Table S1:** The data of this study: the 17 Daisy experiments of `Data/Comprehensive_Summary` (10190 runs, 10041 valid; 20 years, 1980–1999, of Taastrup weather). Means of the valid runs; $Y$ yield (Mg ha$^{-1}$), $I$ irrigation, $D$ drainage at 1 m and $ET$ (mm yr$^{-1}$), $N_L$ nitrate leached (kg N ha$^{-1}$ yr$^{-1}$), $t$ run time of Daisy (s). Every row is one file of the folder, linked.

| Exp. | File | Runs | Valid | Factors varied | $Y$ | $I$ | $D$ | $ET$ | $N_L$ | $t$ |
|---|---|---|---|---|---|---|---|---|---|---|
| E0 | `Baselines` | 10 | 10 | soil | 2.19 | 0 | 55 | 418 | 158 | 231 |
| E1 | `Doses` | 200 | 189 | soil, rate, dose | 5.13 | 198 | 98 | 564 | 79 | 195 |
| E2 | `TriggerPressure` | 200 | 197 | soil, trigger | 4.70 | 169 | 80 | 559 | 90 | 139 |
| E3 | `SensorDepth` | 200 | 200 | soil, sensor depth | 4.94 | 199 | 86 | 583 | 86 | 146 |
| E4 | `Methods` | 100 | 96 | soil, method | 5.01 | 164 | 88 | 542 | 83 | 147 |
| E5 | `SHP Dynamics` | 300 | 300 | soil, method, tillage | 4.13 | 152 | 43 | 579 | 84 | 189 |
| E6 | `Mediterranean Pollution` | 210 | 193 | soil, method, trigger, N rate | 3.64 | 84 | 239 | 620 | 62 | 118 |
| E7 | `Mulching` | 200 | 200 | soil, mulch | 5.29 | 183 | 96 | 555 | 74 | 161 |
| E8 | `Rain Intensity Moisture` | 1000 | 1000 | initial suction, rain scale | 0.68 | 0 | -1 | 282 | -0 | 6 |
| E9 | `Soil Moisture Rain` | 1000 | 987 | soil, initial suction, rain scale | 0.75 | 0 | 19 | 247 | 33 | 7 |
| E10 | `Amount Timing Soil` | 1000 | 954 | soil, rate, dose, window | 4.42 | 139 | 76 | 532 | 97 | 146 |
| E11 | `Temp Evap Rain` | 1000 | 1000 | rain scale, warming, demand | 3.81 | 324 | 18 | 754 | 66 | 151 |
| E12 | `Sensor Trigger Rain` | 1000 | 1000 | trigger, sensor depth, rain scale | 5.88 | 260 | 35 | 670 | 32 | 155 |
| E13 | `N Water Soil` | 1000 | 975 | soil, method, rate, dose, trigger, N rate | 4.00 | 162 | 83 | 547 | 79 | 207 |
| E14 | `Mulch Rain` | 1000 | 1000 | rain scale, mulch | 6.15 | 235 | 43 | 634 | 25 | 173 |
| E15 | `Method Depth Amount` | 1000 | 1000 | method, rate, dose, root depth | 6.02 | 217 | 61 | 619 | 37 | 157 |
| E16 | `Edge Cases` | 770 | 740 | soil, method, rate, dose, trigger, sensor depth, rain scale, warming, N rate, demand | 2.17 | 344 | 185 | 600 | 93 | 224 |

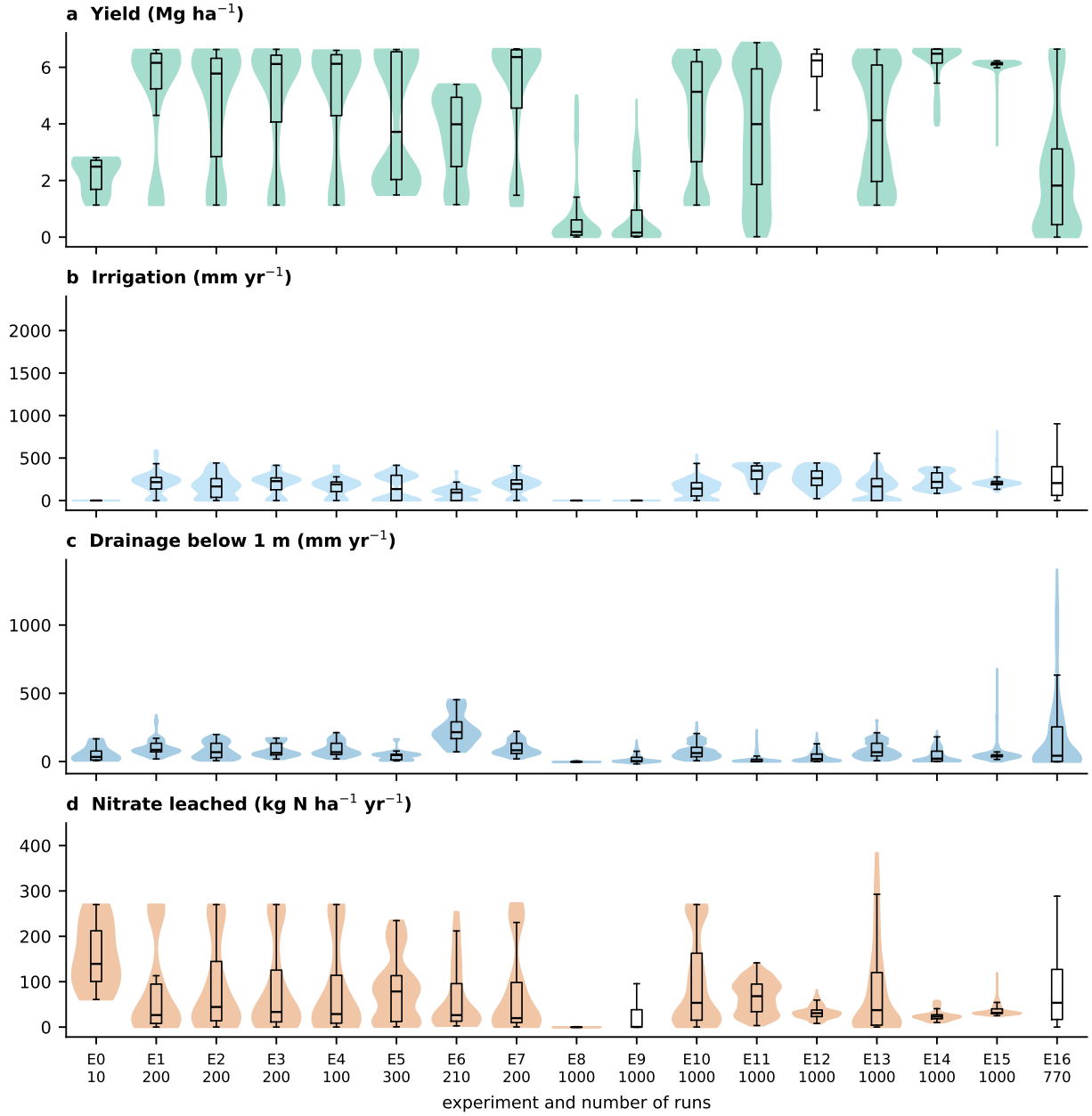



**Figure S1:** Yield, irrigation, drainage and nitrate leaching of the valid runs, by experiment.

## p1_audit_link.json

Source: `Manuscripts/results/p1_audit_link.json`; 46 values. Workbook sheet `R_p1_audit_link`.

| quantity | value |
|---|---|
| E7.S01.psiD_Ent | 0.2975 |
| E7.S01.psiD_Es | 0.3089 |
| E7.S01.share_of_dEnt_in_Es | 0.5618 |
| E7.S01.n | 20 |
| E7.S02.psiD_Ent | 0.6073 |
| E7.S02.psiD_Es | 0.3798 |
| E7.S02.share_of_dEnt_in_Es | 0.4582 |

| quantity | value |
|---|---|
| E7.S02.n | 20 |
| E7.S03.psiD_Ent | 0.7062 |
| E7.S03.psiD_Es | -0.9307 |
| E7.S03.share_of_dEnt_in_Es | -0.3733 |
| E7.S03.n | 20 |
| E7.S04.psiD_Ent | 0.6867 |
| E7.S04.psiD_Es | -0.3497 |
| E7.S04.share_of_dEnt_in_Es | -0.3967 |
| E7.S04.n | 20 |
| E7.S05.psiD_Ent | 0.6628 |
| E7.S05.psiD_Es | 0.157 |
| E7.S05.share_of_dEnt_in_Es | 0.06688 |
| E7.S05.n | 20 |
| E7.S06.psiD_Ent | 0.5212 |
| E7.S06.psiD_Es | 0.4076 |
| E7.S06.share_of_dEnt_in_Es | 0.4787 |
| E7.S06.n | 20 |
| E7.S07.psiD_Ent | 0.3955 |
| E7.S07.psiD_Es | 0.4209 |
| E7.S07.share_of_dEnt_in_Es | 0.5235 |
| E7.S07.n | 20 |
| E7.S08.psiD_Ent | 0.5489 |
| E7.S08.psiD_Es | 0.359 |
| E7.S08.share_of_dEnt_in_Es | 0.3922 |
| E7.S08.n | 20 |
| E7.S09.psiD_Ent | 0.4257 |
| E7.S09.psiD_Es | 0.441 |
| E7.S09.share_of_dEnt_in_Es | 0.5036 |
| E7.S09.n | 20 |
| E7.S10.psiD_Ent | 0.4137 |
| E7.S10.psiD_Es | 0.4541 |
| E7.S10.share_of_dEnt_in_Es | 0.4989 |
| E7.S10.n | 20 |
| E7_summary.median_share_of_dEnt_in_Es | 0.4684 |
| E7_summary.median_abs_diff_psiD | 0.1517 |
| E7_summary.max_abs_diff_psiD | 1.637 |
| E7_summary.n_soils | 10 |
| E7_summary.n_soils_diff_gt_0p1 | 6 |
| E7_summary.n_soils_sign_flip | 2 |

## p1_core.json

Source: `Manuscripts/results/p1_core.json`; 270 values, first 220 shown. Workbook sheet `R_p1_core`.

| quantity | value |
|---|---|
| memory.S01.fE | 0.1451 |
| memory.S01.fT | 0.75 |
| memory.S01.fD | 0.02912 |
| memory.S01.fO | 0.07576 |
| memory.S01.ci.soil_evap_mm | 0.113, 0.2002 |
| memory.S01.ci.transpiration_mm | 0.7102, 0.7721 |
| memory.S01.ci.drainage_1m_mm | 0.02449, 0.03256 |
| memory.S01.ci.oth | 0.06013, 0.09077 |
| memory.S01.capFC | 82.79 |
| memory.S01.cap60 | 112.6 |
| memory.S01.n | 100 |
| memory.S02.fE | 0.1927 |
| memory.S02.fT | 0.4187 |
| memory.S02.fD | 0.265 |
| memory.S02.fO | 0.1236 |
| memory.S02.ci.soil_evap_mm | 0.1604, 0.2324 |
| memory.S02.ci.transpiration_mm | 0.4009, 0.4444 |
| memory.S02.ci.drainage_1m_mm | 0.1982, 0.313 |
| memory.S02.ci.oth | 0.1086, 0.1398 |
| memory.S02.capFC | 104 |
| memory.S02.cap60 | 145.7 |
| memory.S02.n | 100 |
| memory.S03.fE | 0.07403 |
| memory.S03.fT | 0.07982 |
| memory.S03.fD | 0.8455 |
| memory.S03.fO | 0.0006454 |
| memory.S03.ci.soil_evap_mm | 0.06681, 0.08221 |
| memory.S03.ci.transpiration_mm | 0.04333, 0.1216 |
| memory.S03.ci.drainage_1m_mm | 0.7959, 0.89 |
| memory.S03.ci.oth | 0.0002684, 0.001096 |
| memory.S03.capFC | 4.172 |
| memory.S03.cap60 | 9.87 |
| memory.S03.n | 87 |
| memory.S04.fE | 0.09992 |
| memory.S04.fT | 0.1535 |
| memory.S04.fD | 0.7409 |
| memory.S04.fO | 0.005727 |
| memory.S04.ci.soil_evap_mm | 0.093, 0.1086 |
| memory.S04.ci.transpiration_mm | 0.1329, 0.1871 |
| memory.S04.ci.drainage_1m_mm | 0.7008, 0.7655 |
| memory.S04.ci.oth | 0.004453, 0.007196 |
| memory.S04.capFC | 13.19 |
| memory.S04.cap60 | 25.76 |
| memory.S04.n | 100 |
| memory.S05.fE | 0.1277 |
| memory.S05.fT | 0.2043 |
| memory.S05.fD | 0.643 |
| memory.S05.fO | 0.02492 |

| quantity | value |
|---|---|
| memory.S05.ci.soil_evap_mm | 0.1174, 0.1419 |
| memory.S05.ci.transpiration_mm | 0.1839, 0.2358 |
| memory.S05.ci.drainage_1m_mm | 0.5987, 0.6723 |
| memory.S05.ci.oth | 0.02044, 0.0298 |
| memory.S05.capFC | 41.33 |
| memory.S05.cap60 | 71.37 |
| memory.S05.n | 100 |
| memory.S06.fE | 0.04019 |
| memory.S06.fT | 0.5825 |
| memory.S06.fD | 0.2892 |
| memory.S06.fO | 0.08811 |
| memory.S06.ci.soil_evap_mm | 0.02569, 0.05774 |
| memory.S06.ci.transpiration_mm | 0.554, 0.6231 |
| memory.S06.ci.drainage_1m_mm | 0.239, 0.3242 |
| memory.S06.ci.oth | 0.07656, 0.1007 |
| memory.S06.capFC | 131.8 |
| memory.S06.cap60 | 183.1 |
| memory.S06.n | 100 |
| memory.S07.fE | 0.04151 |
| memory.S07.fT | 0.7155 |
| memory.S07.fD | 0.1473 |
| memory.S07.fO | 0.09566 |
| memory.S07.ci.soil_evap_mm | 0.0304, 0.05195 |
| memory.S07.ci.transpiration_mm | 0.6886, 0.7481 |
| memory.S07.ci.drainage_1m_mm | 0.1124, 0.1754 |
| memory.S07.ci.oth | 0.0862, 0.1052 |
| memory.S07.capFC | 208.1 |
| memory.S07.cap60 | 248.9 |
| memory.S07.n | 100 |
| memory.S08.fE | 0.07846 |
| memory.S08.fT | 0.5547 |
| memory.S08.fD | 0.2852 |
| memory.S08.fO | 0.08163 |
| memory.S08.ci.soil_evap_mm | 0.06028, 0.1076 |
| memory.S08.ci.transpiration_mm | 0.5128, 0.593 |
| memory.S08.ci.drainage_1m_mm | 0.2427, 0.3188 |
| memory.S08.ci.oth | 0.07191, 0.09251 |
| memory.S08.capFC | 57.04 |
| memory.S08.cap60 | 84.54 |
| memory.S08.n | 100 |
| memory.S09.fE | 0.02717 |
| memory.S09.fT | 0.7744 |
| memory.S09.fD | 0.08702 |
| memory.S09.fO | 0.1114 |
| memory.S09.ci.soil_evap_mm | 0.01397, 0.03822 |
| memory.S09.ci.transpiration_mm | 0.7541, 0.7978 |
| memory.S09.ci.drainage_1m_mm | 0.06404, 0.1082 |
| memory.S09.ci.oth | 0.1008, 0.1222 |

| quantity | value |
|---|---|
| memory.S09.capFC | 167.5 |
| memory.S09.cap60 | 194.4 |
| memory.S09.n | 100 |
| memory.S10.fE | 0.02489 |
| memory.S10.fT | 0.8416 |
| memory.S10.fD | 0.02197 |
| memory.S10.fO | 0.1115 |
| memory.S10.ci.soil_evap_mm | 0.01344, 0.0364 |
| memory.S10.ci.transpiration_mm | 0.829, 0.8548 |
| memory.S10.ci.drainage_1m_mm | 0.01336, 0.03072 |
| memory.S10.ci.oth | 0.1015, 0.1211 |
| memory.S10.capFC | 173 |
| memory.S10.cap60 | 190.1 |
| memory.S10.n | 100 |
| rec16.S01.psiD | 0.2946 |
| rec16.S01.psiI | 0.6189 |
| rec16.S01.psiT | 0.1113 |
| rec16.S01.psiS | 0.02473 |
| rec16.S01.dE | 218.5 |
| rec16.S01.ci.psiD | 0.1943, 0.3918 |
| rec16.S01.ci.psiI | 0.448, 0.7595 |
| rec16.S01.ci.psiT | -0.04115, 0.3197 |
| rec16.S01.ci.psiS | 0.00506, 0.04393 |
| rec16.S01.ci.dE | 179.2, 257.9 |
| rec16.S01.n | 32 |
| rec16.S02.psiD | 0.193 |
| rec16.S02.psiI | 0.6578 |
| rec16.S02.psiT | 0.157 |
| rec16.S02.psiS | 0.007809 |
| rec16.S02.dE | 143.6 |
| rec16.S02.ci.psiD | 0.1389, 0.2439 |
| rec16.S02.ci.psiI | 0.3244, 0.858 |
| rec16.S02.ci.psiT | -0.02457, 0.4575 |
| rec16.S02.ci.psiS | 0.004907, 0.0132 |
| rec16.S02.ci.dE | 99.58, 201.3 |
| rec16.S02.n | 13 |
| rec16.S03.psiD | 1.21 |
| rec16.S03.psiI | -0.3839 |
| rec16.S03.psiT | 0.1785 |
| rec16.S03.psiS | 0.004424 |
| rec16.S03.dE | 99.64 |
| rec16.S03.ci.psiD | 0.955, 1.46 |
| rec16.S03.ci.psiI | -0.8056, -0.05112 |
| rec16.S03.ci.psiT | 0.01523, 0.4111 |
| rec16.S03.ci.psiS | -0.001305, 0.01098 |
| rec16.S03.ci.dE | 86.72, 112.8 |
| rec16.S03.n | 32 |
| rec16.S04.psiD | 0.7993 |

| quantity | value |
|---|---|
| rec16.S04.psiI | 0.2189 |
| rec16.S04.psiT | -0.007351 |
| rec16.S04.psiS | 0.01086 |
| rec16.S04.dE | 115 |
| rec16.S04.ci.psiD | 0.6691, 0.9158 |
| rec16.S04.ci.psiI | 0.1352, 0.3027 |
| rec16.S04.ci.psiT | -0.1054, 0.1089 |
| rec16.S04.ci.psiS | 0.005401, 0.01805 |
| rec16.S04.ci.dE | 99.12, 129.1 |
| rec16.S04.n | 32 |
| rec16.S05.psiD | 0.5821 |
| rec16.S05.psiI | 0.4146 |
| rec16.S05.psiT | 0.01543 |
| rec16.S05.psiS | 0.0121 |
| rec16.S05.dE | 141.1 |
| rec16.S05.ci.psiD | 0.4566, 0.6966 |
| rec16.S05.ci.psiI | 0.2391, 0.5755 |
| rec16.S05.ci.psiT | -0.1053, 0.177 |
| rec16.S05.ci.psiS | 0.003533, 0.02239 |
| rec16.S05.ci.dE | 115.9, 167.7 |
| rec16.S05.n | 32 |
| rec16.S06.psiD | 0.469 |
| rec16.S06.psiI | 0.4334 |
| rec16.S06.psiT | 0.1068 |
| rec16.S06.psiS | 0.009196 |
| rec16.S06.dE | 191.5 |
| rec16.S06.ci.psiD | 0.3626, 0.5992 |
| rec16.S06.ci.psiI | 0.1834, 0.607 |
| rec16.S06.ci.psiT | -0.0311, 0.3117 |
| rec16.S06.ci.psiS | -0.003642, 0.02676 |
| rec16.S06.ci.dE | 139.4, 247.3 |
| rec16.S06.n | 32 |
| rec16.S07.psiD | 0.3625 |
| rec16.S07.psiI | 0.5278 |
| rec16.S07.psiT | 0.1168 |
| rec16.S07.psiS | 0.007042 |
| rec16.S07.dE | 217.6 |
| rec16.S07.ci.psiD | 0.2597, 0.4852 |
| rec16.S07.ci.psiI | 0.2777, 0.7039 |
| rec16.S07.ci.psiT | -0.03777, 0.348 |
| rec16.S07.ci.psiS | -0.009443, 0.02719 |
| rec16.S07.ci.dE | 168.3, 271.5 |
| rec16.S07.n | 32 |
| rec16.S08.psiD | 0.533 |
| rec16.S08.psiI | 0.3857 |
| rec16.S08.psiT | 0.09111 |
| rec16.S08.psiS | 0.009891 |
| rec16.S08.dE | 152.5 |

| quantity | value |
|---|---|
| rec16.S08.ci.psiD | 0.4093, 0.6608 |
| rec16.S08.ci.psiI | 0.1227, 0.5953 |
| rec16.S08.ci.psiT | -0.07816, 0.3433 |
| rec16.S08.ci.psiS | -0.005323, 0.02717 |
| rec16.S08.ci.dE | 114.9, 190.4 |
| rec16.S08.n | 32 |
| rec16.S09.psiD | 0.3117 |
| rec16.S09.psiI | 0.645 |
| rec16.S09.psiT | 0.04777 |
| rec16.S09.psiS | 0.004532 |
| rec16.S09.dE | 217.4 |
| rec16.S09.ci.psiD | 0.1979, 0.4445 |
| rec16.S09.ci.psiI | 0.3863, 0.806 |
| rec16.S09.ci.psiT | -0.07744, 0.2568 |
| rec16.S09.ci.psiS | -0.01091, 0.02484 |
| rec16.S09.ci.dE | 165.6, 267.7 |
| rec16.S09.n | 32 |
| rec16.S10.psiD | 0.3103 |
| rec16.S10.psiI | 0.838 |
| rec16.S10.psiT | -0.01707 |
| rec16.S10.psiS | 0.1312 |
| rec16.S10.dE | 222.1 |
| rec16.S10.ci.psiD | 0.217, 0.4126 |
| rec16.S10.ci.psiI | 0.6104, 1.033 |
| rec16.S10.ci.psiT | -0.1507, 0.1387 |
| rec16.S10.ci.psiS | 0.05407, 0.1948 |
| rec16.S10.ci.dE | 176.8, 272.6 |
| rec16.S10.n | 32 |

## p1_ens_structure.json

Source: `Manuscripts/results/p1_ens_structure.json`; 50 values. Workbook sheet `R_p1_ens_structure`.

| quantity | value |
|---|---|
| dE.S01 | 54.3 |
| dE.S02 | 28.7 |
| dE.S03 | 3.4 |
| dE.S04 | 6.9 |
| dE.S05 | 17 |
| dE.S06 | 32.8 |
| dE.S07 | 45.3 |
| dE.S08 | 25.5 |
| dE.S09 | 41.2 |
| dE.S10 | 44.8 |

| quantity | value |
|---|---|
| dD.S01 | 16.5 |
| dD.S02 | 20.8 |
| dD.S03 | 6.1 |
| dD.S04 | 7.9 |
| dD.S05 | 9.5 |
| dD.S06 | 19.5 |
| dD.S07 | 25.4 |
| dD.S08 | 15.4 |
| dD.S09 | 25.8 |
| dD.S10 | 27.1 |
| dI.S01 | 46.6 |
| dI.S02 | 10.4 |
| dI.S03 | 0 |
| dI.S04 | -0.4 |
| dI.S05 | -5.5 |
| dI.S06 | 18.1 |
| dI.S07 | 45.1 |
| dI.S08 | 9.2 |
| dI.S09 | 42.8 |
| dI.S10 | 49.1 |
| tillage_offset.S01 | 0.27 |
| tillage_offset.S02 | -1.82 |
| tillage_offset.S03 | 5.34 |
| tillage_offset.S04 | 3.69 |
| tillage_offset.S05 | 1.76 |
| tillage_offset.S06 | -1.32 |
| tillage_offset.S07 | -0.25 |
| tillage_offset.S08 | -3.69 |
| tillage_offset.S09 | -2.79 |
| tillage_offset.S10 | -1.7 |
| tc_effect.S01 | 0 |
| tc_effect.S02 | 0 |
| tc_effect.S03 | 0 |
| tc_effect.S04 | 0 |
| tc_effect.S05 | 0 |
| tc_effect.S06 | 1.09 |
| tc_effect.S07 | 0.23 |
| tc_effect.S08 | 2.54 |
| tc_effect.S09 | 2.06 |
| tc_effect.S10 | 1.48 |

## p1_loso_summary.json

Source: `Manuscripts/results/p1_loso_summary.json`; 5 values. Workbook sheet `R_p1_loso_summary`.

| quantity | value |
|---|---|
| TAW | 0.1825 |
| $S_e$(100) | 0.1358 |
| $n$ | 0.09716 |
| $/log K_s$* | 0.05811 |
| mean only | 0.3178 |

## p1_mech.json

Source: `Manuscripts/results/p1_mech.json`; 5 values. Workbook sheet `R_p1_mech`.

| quantity | value |
|---|---|
| hstar | 6.712 |
| q | 15.41 |
| pred | 0.8355, 0.7637, 0.6336, 0.3368, 0.1148, 0.2433, 0.08474, 0.02309 |
| loo | 0.8219, 0.7791, 0.63, 0.3715, 0.1108, 0.2334, 0.08459, 0.02309 |
| soils | S03, S04, S05, S06, S07, S08, S09, S10 |

## p1_modifiers.json

Source: `Manuscripts/results/p1_modifiers.json`; 20 values. Workbook sheet `R_p1_modifiers`.

| quantity | value |
|---|---|
| fP.mean | 0.4193 |
| fP.pos | 9 |
| fP.n | 9 |
| fP.p | 0.003906 |
| fT.mean | -0.1612 |
| fT.pos | 0 |
| fT.n | 10 |
| fT.p | 0.001953 |
| fh.mean | 0.1364 |
| fh.pos | 7 |
| fh.n | 10 |
| fh.p | 0.1602 |
| fD.mean | -0.04023 |
| fD.pos | 2 |
| fD.n | 10 |
| fD.p | 0.3223 |
| fN.mean | -0.008285 |
| fN.pos | 2 |
| fN.n | 10 |
| fN.p | 0.06445 |

## p1_rank.json

Source: `Manuscripts/results/p1_rank.json`; 20 values. Workbook sheet `R_p1_rank`.

| quantity | value |
|---|---|
| $n$ vs $/phi_D$.rho | 0.9515 |
| $n$ vs $/phi_D$.lo | 0.6883 |
| $n$ vs $/phi_D$.hi | 1 |
| $n$ vs $/phi_D$.p | 0.00015 |
| $/phi_D$ vs $/psi_D$(E16).rho | 0.8545 |
| $/phi_D$ vs $/psi_D$(E16).lo | 0.3703 |
| $/phi_D$ vs $/psi_D$(E16).hi | 1 |
| $/phi_D$ vs $/psi_D$(E16).p | 0.00195 |
| $/phi_D$ vs $/psi_D$(E7).rho | 0.8909 |
| $/phi_D$ vs $/psi_D$(E7).lo | 0.5472 |
| $/phi_D$ vs $/psi_D$(E7).hi | 1 |
| $/phi_D$ vs $/psi_D$(E7).p | 0.00135 |
| $n$ vs $/psi_D$(E16).rho | 0.697 |
| $n$ vs $/psi_D$(E16).lo | -0.0566 |
| $n$ vs $/psi_D$(E16).hi | 1 |
| $n$ vs $/psi_D$(E16).p | 0.03185 |
| $n$ vs $/psi_D$(E7).rho | 0.8909 |
| $n$ vs $/psi_D$(E7).lo | 0.4717 |
| $n$ vs $/psi_D$(E7).hi | 1 |
| $n$ vs $/psi_D$(E7).p | 0.001 |

## p1_rate_law.json

Source: `Manuscripts/results/p1_rate_law.json`; 17 values. Workbook sheet `R_p1_rate_law`.

| quantity | value |
|---|---|
| q_star_cm_d | 71.46 |
| R2 | 0.9715 |
| boot_ci95 | 58.38, 105.2 |
| n_boot | 4999 |
| jackknife_range | 65.83, 79.97 |
| loso_rmse_rate_law | 0.05811 |
| loso_8.mean_only | 0.3384 |
| loso_8.TAW | 0.08835 |
| loso_8.Se100 | 0.06407 |
| loso_8.n | 0.1004 |
| loso_8.log10Ks_linear | 0.06183 |
| loso_10.mean_only | 0.3178 |
| loso_10.TAW | 0.1825 |
| loso_10.Se100 | 0.1358 |
| loso_10.n | 0.09716 |

| quantity | value |
|---|---|
| loo_pred | 0.9116, 0.8418, 0.5702, 0.2502, 0.07804, 0.3136, 0.08007, 0.02297 |
| soils | S03, S04, S05, S06, S07, S08, S09, S10 |

## p1_table.csv

Source: `Manuscripts/results/p1_table.csv`; 10 rows. Workbook sheet `R_p1_table`.

| Unnamed: 0 | phiD_E9 | phiT_E9 | phiE_E9 | capFC | psiD_E16 | psiI_E16 | psiT_E16 | dE16 |
|---|---|---|---|---|---|---|---|---|
| S01 | 0.02912 | 0.75 | 0.1451 | 82.79 | 0.2946 | 0.6189 | 0.1113 | 218.5 |
| S02 | 0.265 | 0.4187 | 0.1927 | 104 | 0.193 | 0.6578 | 0.157 | 143.6 |
| S03 | 0.8455 | 0.07982 | 0.07403 | 4.172 | 1.21 | -0.3839 | 0.1785 | 99.64 |
| S04 | 0.7409 | 0.1535 | 0.09992 | 13.19 | 0.7993 | 0.2189 | -0.007351 | 115 |
| S05 | 0.643 | 0.2043 | 0.1277 | 41.33 | 0.5821 | 0.4146 | 0.01543 | 141.1 |
| S06 | 0.2892 | 0.5825 | 0.04019 | 131.8 | 0.469 | 0.4334 | 0.1068 | 191.5 |
| S07 | 0.1473 | 0.7155 | 0.04151 | 208.1 | 0.3625 | 0.5278 | 0.1168 | 217.6 |
| S08 | 0.2852 | 0.5547 | 0.07846 | 57.04 | 0.533 | 0.3857 | 0.09111 | 152.5 |
| S09 | 0.08702 | 0.7744 | 0.02717 | 167.5 | 0.3117 | 0.645 | 0.04777 | 217.4 |
| S10 | 0.02197 | 0.8416 | 0.02489 | 173 | 0.3103 | 0.838 | -0.01707 | 222.1 |

## p1_table2.csv

Source: `Manuscripts/results/p1_table2.csv`; 10 rows. Workbook sheet `R_p1_table2`.

| Unnamed: 0 | phiD_E9 | phiT_E9 | phiE_E9 | capFC | psiD_E16 | psiI_E16 | psiT_E16 | dE16 |
|---|---|---|---|---|---|---|---|---|
| S01 | 0.02912 | 0.75 | 0.1451 | 82.79 | 0.2946 | 0.6189 | 0.1113 | 218.5 |
| S02 | 0.265 | 0.4187 | 0.1927 | 104 | 0.193 | 0.6578 | 0.157 | 143.6 |
| S03 | 0.8455 | 0.07982 | 0.07403 | 4.172 | 1.21 | -0.3839 | 0.1785 | 99.64 |
| S04 | 0.7409 | 0.1535 | 0.09992 | 13.19 | 0.7993 | 0.2189 | -0.007351 | 115 |
| S05 | 0.643 | 0.2043 | 0.1277 | 41.33 | 0.5821 | 0.4146 | 0.01543 | 141.1 |
| S06 | 0.2892 | 0.5825 | 0.04019 | 131.8 | 0.469 | 0.4334 | 0.1068 | 191.5 |
| S07 | 0.1473 | 0.7155 | 0.04151 | 208.1 | 0.3625 | 0.5278 | 0.1168 | 217.6 |
| S08 | 0.2852 | 0.5547 | 0.07846 | 57.04 | 0.533 | 0.3857 | 0.09111 | 152.5 |
| S09 | 0.08702 | 0.7744 | 0.02717 | 167.5 | 0.3117 | 0.645 | 0.04777 | 217.4 |
| S10 | 0.02197 | 0.8416 | 0.02489 | 173 | 0.3103 | 0.838 | -0.01707 | 222.1 |

## p2_R.json

Source: `Manuscripts/results/p2_R.json`; 27 values. Workbook sheet `R_p2_R`.

| quantity | value |
|---|---|
| R_model.terms | lnI, n, cap, dT |

| quantity | value |
|---|---|
| R_model.coef.const | 3.355 |
| R_model.coef.lnI | 0.1944 |
| R_model.coef.n | 0.4917 |
| R_model.coef.cap | 0.1454 |
| R_model.coef.dT | 0.2383 |
| R_model.se.const | 0.02376 |
| R_model.se.lnI | 0.001882 |
| R_model.se.n | 0.01289 |
| R_model.se.cap | 0.004217 |
| R_model.se.dT | 0.00723 |
| R_model.r2 | 0.6111 |
| R_model.path[0] | lnI, 0.5509 |
| R_model.path[1] | n, 0.5356 |
| R_model.path[2] | cap, 0.5169 |
| R_model.path[3] | dT, 0.5137 |
| R_model.loeo_rmse_ln | 0.5137 |
| R_sparse.R2 | 0.3028 |
| R_sparse.rmse | 80.54 |
| R_sparse.R2ln | 0.4092 |
| R_GBM_all.R2 | 0.4956 |
| R_GBM_all.rmse | 68.51 |
| R_GBM_all.R2ln | 0.4268 |
| R_GBM_sel.R2 | 0.5095 |
| R_GBM_sel.rmse | 67.56 |
| R_GBM_sel.R2ln | 0.4665 |
| R_sparse_LOSO_R2ln | 0.4275 |

## p2_a.json

Source: `Manuscripts/results/p2_a.json`; 46 values. Workbook sheet `R_p2_a`.

| quantity | value |
|---|---|
| svd.sv | 240.3, 82.6, 57.95, 39.37, 21.84, 18.17, 5.461 |
| svd.null[0].k | 1 |
| svd.null[0].sv | 5.461 |
| svd.null[0].vec.rain_mm | 0.8956 |
| svd.null[0].vec.irrigation_mm | 0.8343 |
| svd.null[0].vec.ET_mm | -1 |
| svd.null[0].vec.transpiration_mm | 0.3675 |
| svd.null[0].vec.soil_evap_mm | 0.1207 |
| svd.null[0].vec.canopy_evap_mm | -0.1216 |
| svd.null[0].vec.drainage_1m_mm | -0.9325 |
| svd.null[1].k | 2 |
| svd.null[1].sv | 18.17 |
| svd.null[1].vec.rain_mm | -0.3585 |
| svd.null[1].vec.irrigation_mm | -0.2111 |

| quantity | value |
|---|---|
| svd.null[1].vec.ET_mm | -0.1925 |
| svd.null[1].vec.transpiration_mm | 0.7876 |
| svd.null[1].vec.soil_evap_mm | 0.4982 |
| svd.null[1].vec.canopy_evap_mm | 1 |
| svd.null[1].vec.drainage_1m_mm | 0.1799 |
| svd.null[2].k | 3 |
| svd.null[2].sv | 21.84 |
| svd.null[2].vec.rain_mm | -0.03927 |
| svd.null[2].vec.irrigation_mm | 0.03899 |
| svd.null[2].vec.ET_mm | -0.02946 |
| svd.null[2].vec.transpiration_mm | -0.3726 |
| svd.null[2].vec.soil_evap_mm | 0.2027 |
| svd.null[2].vec.canopy_evap_mm | 1 |
| svd.null[2].vec.drainage_1m_mm | -0.0192 |
| balance (W-ET-D) | 47.83, 5.435, 81.74 |
| ET partition (ET-T-Es-Ec) | 190, 129, 334.7 |
| clrpca | 0.8858, 0.07317, 0.03278, 0.008299, 0 |
| clr_load.transpiration_mm.0 | -0.218 |
| clr_load.transpiration_mm.1 | 0.035 |
| clr_load.transpiration_mm.2 | -0.22 |
| clr_load.soil_evap_mm.0 | -0.334 |
| clr_load.soil_evap_mm.1 | -0.557 |
| clr_load.soil_evap_mm.2 | -0.449 |
| clr_load.canopy_evap_mm.0 | -0.034 |
| clr_load.canopy_evap_mm.1 | 0.801 |
| clr_load.canopy_evap_mm.2 | -0.21 |
| clr_load.R.0 | -0.285 |
| clr_load.R.1 | -0.079 |
| clr_load.R.2 | 0.839 |
| clr_load.drainage_1m_mm.0 | 0.871 |
| clr_load.drainage_1m_mm.1 | -0.2 |
| clr_load.drainage_1m_mm.2 | 0.04 |

## p2_dpf.json

Source: `Manuscripts/results/p2_dpf.json`; 38 values. Workbook sheet `R_p2_dpf`.

| quantity | value |
|---|---|
| exogenous.terms | lnW/_TAW, lnalpha, Ifrac, dT |
| exogenous.hist[0] | lnW/_TAW, 0.1926 |
| exogenous.hist[1] | lnalpha, 0.2601 |
| exogenous.hist[2] | Ifrac, 0.422 |
| exogenous.hist[3] | dT, 0.4334 |
| exogenous.loeo | 0.4334 |
| exogenous.loso | 0.2171 |
| exogenous.r2 | 0.5752 |

| quantity | value |
|---|---|
| exogenous.coef.const | -12.1 |
| exogenous.coef.lnW_TAW | 4.411 |
| exogenous.coef.lnalpha | -1.214 |
| exogenous.coef.Ifrac | -5.565 |
| exogenous.coef.dT | -0.9978 |
| exogenous.se.const | 0.1018 |
| exogenous.se.lnW_TAW | 0.03957 |
| exogenous.se.lnalpha | 0.0273 |
| exogenous.se.Ifrac | 0.1049 |
| exogenous.se.dT | 0.03043 |
| exogenous.gbm_loeo | 0.4975 |
| with_partition.terms | lnW/_TAW, lnalpha, Ifrac, dT |
| with_partition.hist[0] | lnW/_TAW, 0.1926 |
| with_partition.hist[1] | lnalpha, 0.2601 |
| with_partition.hist[2] | Ifrac, 0.422 |
| with_partition.hist[3] | dT, 0.4334 |
| with_partition.loeo | 0.4334 |
| with_partition.loso | 0.2171 |
| with_partition.r2 | 0.5752 |
| with_partition.coef.const | -12.1 |
| with_partition.coef.lnW_TAW | 4.411 |
| with_partition.coef.lnalpha | -1.214 |
| with_partition.coef.Ifrac | -5.565 |
| with_partition.coef.dT | -0.9978 |
| with_partition.se.const | 0.1018 |
| with_partition.se.lnW_TAW | 0.03957 |
| with_partition.se.lnalpha | 0.0273 |
| with_partition.se.Ifrac | 0.1049 |
| with_partition.se.dT | 0.03043 |
| with_partition.gbm_loeo | 0.5906 |

## p2_et.json

Source: `Manuscripts/results/p2_et.json`; 90 values. Workbook sheet `R_p2_et`.

| quantity | value |
|---|---|
| lnET.terms | lnW, n, lnI1 |
| lnET.hist[0] | lnW, 0.8008 |
| lnET.hist[1] | n, 0.8947 |
| lnET.hist[2] | lnI1, 0.8982 |
| lnET.loeo | 0.8982 |
| lnET.loso | 0.9037 |
| lnET.r2 | 0.9082 |
| lnET.coef.const | 2.503 |
| lnET.coef.lnW | 0.6458 |
| lnET.coef.n | -0.2992 |

| quantity | value |
|---|---|
| lnET.coef.lnI1 | 0.02932 |
| lnET.se.const | 0.02691 |
| lnET.se.lnW | 0.005084 |
| lnET.se.n | 0.004524 |
| lnET.se.lnI1 | 0.001125 |
| lnET.gbm_loeo | 0.8789 |
| lnET.gbm_loso | 0.8936 |
| lnET.rmse_loeo | 0.143 |
| lnT.terms | lnW, n |
| lnT.hist[0] | lnW, 0.5805 |
| lnT.hist[1] | n, 0.6409 |
| lnT.loeo | 0.6409 |
| lnT.loso | 0.6524 |
| lnT.r2 | 0.6712 |
| lnT.coef.const | -1.329 |
| lnT.coef.lnW | 1.127 |
| lnT.coef.n | -0.5007 |
| lnT.se.const | 0.06105 |
| lnT.se.lnW | 0.00874 |
| lnT.se.n | 0.01243 |
| lnT.gbm_loeo | 0.6415 |
| lnT.gbm_loso | 0.6592 |
| lnT.rmse_loeo | 0.4588 |
| lnEs.terms | n, mulch, lnrain |
| lnEs.hist[0] | n, 0.07031 |
| lnEs.hist[1] | mulch, 0.2369 |
| lnEs.hist[2] | lnrain, 0.3084 |
| lnEs.loeo | 0.3084 |
| lnEs.loso | 0.3484 |
| lnEs.r2 | 0.4363 |
| lnEs.coef.const | 3.271 |
| lnEs.coef.n | -0.7698 |
| lnEs.coef.mulch | -0.3804 |
| lnEs.coef.lnrain | 0.4616 |
| lnEs.se.const | 0.06683 |
| lnEs.se.n | 0.01217 |
| lnEs.se.mulch | 0.006953 |
| lnEs.se.lnrain | 0.01089 |
| lnEs.gbm_loeo | 0.5468 |
| lnEs.gbm_loso | 0.4758 |
| lnEs.rmse_loeo | 0.4803 |
| lnEc.terms | lnW, lnrain, Ifrac |
| lnEc.hist[0] | lnW, 0.6408 |
| lnEc.hist[1] | lnrain, 0.6513 |
| lnEc.hist[2] | Ifrac, 0.6677 |
| lnEc.loeo | 0.6677 |
| lnEc.loso | 0.6695 |
| lnEc.r2 | 0.7035 |

| quantity | value |
|---|---|
| lnEc.coef.const | -13.32 |
| lnEc.coef.lnW | -0.4701 |
| lnEc.coef.lnrain | 3.157 |
| lnEc.coef.Ifrac | 4.24 |
| lnEc.se.const | 0.1448 |
| lnEc.se.lnW | 0.1351 |
| lnEc.se.lnrain | 0.1263 |
| lnEc.se.Ifrac | 0.2092 |
| lnEc.gbm_loeo | 0.6914 |
| lnEc.gbm_loso | 0.8059 |
| lnEc.rmse_loeo | 0.9389 |
| lnY.terms | lnW, lnI1, lnrain, mulch |
| lnY.hist[0] | lnW, 0.5314 |
| lnY.hist[1] | lnI1, 0.5492 |
| lnY.hist[2] | lnrain, 0.5992 |
| lnY.hist[3] | mulch, 0.604 |
| lnY.loeo | 0.604 |
| lnY.loso | 0.6169 |
| lnY.r2 | 0.6354 |
| lnY.coef.const | -9.681 |
| lnY.coef.lnW | -0.162 |
| lnY.coef.lnI1 | 0.3226 |
| lnY.coef.lnrain | 1.716 |
| lnY.coef.mulch | 0.1489 |
| lnY.se.const | 0.1806 |
| lnY.se.lnW | 0.05911 |
| lnY.se.lnI1 | 0.008161 |
| lnY.se.lnrain | 0.04756 |
| lnY.se.mulch | 0.01476 |
| lnY.gbm_loeo | 0.6501 |
| lnY.gbm_loso | 0.7425 |
| lnY.rmse_loeo | 0.9531 |

## p2_etlaw.json

Source: `Manuscripts/results/p2_etlaw.json`; 31 values. Workbook sheet `R_p2_etlaw`.

| quantity | value |
|---|---|
| sparse_loeo | 0.8982 |
| sparse_loso | 0.9037 |
| gbm_loeo | 0.876 |
| gbm_loso | 0.8812 |
| coef_mean | 2.508, 0.6448, -0.2984, 0.02945 |
| coef_sd | 0.0771, 0.01676, 0.01733, 0.002268 |
| coefS_mean | 2.498, 0.6465, -0.2993, 0.02912 |
| coefS_sd | 0.07797, 0.0124, 0.007435, 0.002257 |

| quantity | value |
|---|---|
| rmse_mm_loeo | 68.82 |
| mape_loeo | 9.179 |
| by_exp[0] | E0/_Baselines, 10, 5.834, -25.17 |
| by_exp[1] | E10/_Amount/_Timing/_Soil, 954, 3.256, 1.285 |
| by_exp[2] | E11/_Temp/_Evap/_Rain, 1000, 8.999, -62.58 |
| by_exp[3] | E12/_Sensor/_Trigger/_Rain, 1000, 3.268, -15.24 |
| by_exp[4] | E13/_N/_Water/_Soil, 975, 3.532, -3.65 |
| by_exp[5] | E14/_Mulch/_Rain, 1000, 4.024, 7.484 |
| by_exp[6] | E15/_Method/_Depth/_Amount, 1000, 4.978, 28.96 |
| by_exp[7] | E16/_Edge/_Cases, 740, 16.6, 34.49 |
| by_exp[8] | E1/_Doses, 189, 3.76, 9.833 |
| by_exp[9] | E2/_TriggerPressure, 197, 3.002, -6.6 |
| by_exp[10] | E3/_SensorDepth, 200, 2.648, -8.047 |
| by_exp[11] | E4/_Methods, 96, 3.557, 9.425 |
| by_exp[12] | E5/_SHP/_Dynamics, 300, 11.7, -63.8 |
| by_exp[13] | E6/_Mediterranean/_Pollution, 193, 8.457, 41.52 |
| by_exp[14] | E7/_Mulching, 200, 4.66, 10.62 |
| by_exp[15] | E8/_Rain/_Intensity/_Moisture, 1000, 16.72, -11.34 |
| by_exp[16] | E9/_Soil/_Moisture/_Rain, 987, 27.4, 12.89 |
| closureD.r2 | 0.6877 |
| closureD.rmse | 63.71 |
| closureD.n | 8054 |
| closureD.negshare | 0.04817 |

## p2_laws.json

Source: `Manuscripts/results/p2_laws.json`; 12 values. Workbook sheet `R_p2_laws`.

| quantity | value |
|---|---|
| svd4.vec | 1.037, 0.9575, -1, -1.036 |
| svd4.lo | 1.031, 0.9396, -1, -1.052 |
| svd4.hi | 1.045, 0.9738, -1, -1.023 |
| svd4.sv | 176.6, 75.65, 56.89, 5.92 |
| svd4_perm_minSV | 33.33 |
| svd4_noE89.vec | 1.021, 0.9688, -1, -1.019 |
| svd4_noE89.sv | 158.9, 67.04, 49.48, 3.007 |
| closure.all_median_abs | 0.009134 |
| closure.p95 | 0.3515 |
| closure.noE89_median | 0.006472 |
| closure.noE89_p95 | 0.02268 |
| closure.E89_median | 0.1643 |

## p2_laws2.json

Source: `Manuscripts/results/p2_laws2.json`; 81 values. Workbook sheet `R_p2_laws2`.

| quantity | value |
|---|---|
| comp_lattice[0].exp | E11_Temp_Evap_Rain |
| comp_lattice[0].n | 1000 |
| comp_lattice[0].irrigation_mm_slope | -0.4193 |
| comp_lattice[0].irrigation_mm_se | 0.01881 |
| comp_lattice[0].ET_mm_slope | 0.4302 |
| comp_lattice[0].ET_mm_se | 0.02525 |
| comp_lattice[0].drainage_1m_mm_slope | 0.1333 |
| comp_lattice[0].drainage_1m_mm_se | 0.006955 |
| comp_lattice[0].W_slope | 0.5807 |
| comp_lattice[0].W_se | 0.01881 |
| comp_lattice[1].exp | E12_Sensor_Trigger_Rain |
| comp_lattice[1].n | 1000 |
| comp_lattice[1].irrigation_mm_slope | -0.673 |
| comp_lattice[1].irrigation_mm_se | 0.01357 |
| comp_lattice[1].ET_mm_slope | 0.07769 |
| comp_lattice[1].ET_mm_se | 0.009059 |
| comp_lattice[1].drainage_1m_mm_slope | 0.2316 |
| comp_lattice[1].drainage_1m_mm_se | 0.006128 |
| comp_lattice[1].W_slope | 0.327 |
| comp_lattice[1].W_se | 0.01357 |
| comp_lattice[2].exp | E14_Mulch_Rain |
| comp_lattice[2].n | 1000 |
| comp_lattice[2].irrigation_mm_slope | -0.6656 |
| comp_lattice[2].irrigation_mm_se | 0.006611 |
| comp_lattice[2].ET_mm_slope | -0.01901 |
| comp_lattice[2].ET_mm_se | 0.009262 |
| comp_lattice[2].drainage_1m_mm_slope | 0.3204 |
| comp_lattice[2].drainage_1m_mm_se | 0.005513 |
| comp_lattice[2].W_slope | 0.3344 |
| comp_lattice[2].W_se | 0.006611 |
| comp16.S01.kappa | 0.3033 |
| comp16.S01.dET | 0.5118 |
| comp16.S01.dD | 0.186 |
| comp16.S01.n | 32 |
| comp16.S02.kappa | 0.1641 |
| comp16.S02.dET | 0.346 |
| comp16.S02.dD | 0.4954 |
| comp16.S02.n | 8 |
| comp16.S03.kappa | -0.0141 |
| comp16.S03.dET | 0.4732 |
| comp16.S03.dD | 0.5502 |
| comp16.S03.n | 32 |
| comp16.S04.kappa | 0.08366 |

| quantity | value |
|---|---|
| comp16.S04.dET | 0.4733 |
| comp16.S04.dD | 0.4532 |
| comp16.S04.n | 32 |
| comp16.S05.kappa | 0.1021 |
| comp16.S05.dET | 0.5515 |
| comp16.S05.dD | 0.3576 |
| comp16.S05.n | 32 |
| comp16.S06.kappa | 0.1082 |
| comp16.S06.dET | 0.6193 |
| comp16.S06.dD | 0.2813 |
| comp16.S06.n | 32 |
| comp16.S07.kappa | 0.2158 |
| comp16.S07.dET | 0.5423 |
| comp16.S07.dD | 0.2464 |
| comp16.S07.n | 32 |
| comp16.S08.kappa | 0.08991 |
| comp16.S08.dET | 0.6131 |
| comp16.S08.dD | 0.3076 |
| comp16.S08.n | 32 |
| comp16.S09.kappa | 0.2382 |
| comp16.S09.dET | 0.5154 |
| comp16.S09.dD | 0.2529 |
| comp16.S09.n | 32 |
| comp16.S10.kappa | 0.3016 |
| comp16.S10.dET | 0.4768 |
| comp16.S10.dD | 0.2369 |
| comp16.S10.n | 32 |
| kappa16_pool.est | 0.1589 |
| kappa16_pool.lo | 0.1336 |
| kappa16_pool.hi | 0.1813 |
| kap_by.fEp.0.3 | 0.1927 |
| kap_by.fEp.1.2 | 0.1232 |
| kap_by.fT.2.0 | 0.188 |
| kap_by.fT.6.0 | 0.1289 |
| kap_by.fh.50.0 | 0.211 |
| kap_by.fh.2000.0 | 0.1067 |
| kap_by.fD.1.2 | 0.0791 |
| kap_by.fD.30.0 | 0.2408 |

## p2_nested.json

Source: `Manuscripts/results/p2_nested.json`; 18 values. Workbook sheet `R_p2_nested`.

| quantity | value |
|---|---|
| r2 | 0.8946 |
| sets[0] | lnW, n, lnI1 |

| quantity | value |
|---|---|
| sets[1] | lnW, n, lnI1 |
| sets[2] | lnW, n, lnI1 |
| sets[3] | lnW, n, lnI1 |
| sets[4] | lnW, n, lnI1 |
| sets[5] | lnW, n, lnI1 |
| sets[6] | lnW, n, lnI1 |
| sets[7] | lnW, n |
| sets[8] | lnW, n, lnI1 |
| sets[9] | lnW, n, lnI1 |
| sets[10] | lnW, n, lnI1 |
| sets[11] | lnW, n, lnI1 |
| sets[12] | lnW, n, lnI1 |
| sets[13] | lnW, n |
| sets[14] | lnW, n, lnI1 |
| sets[15] | lnW, n, lnI1 |
| sets[16] | lnW, n, lnI1 |

## p2_pca.json

Source: `Manuscripts/results/p2_pca.json`; 4 values. Workbook sheet `R_p2_pca`.

| quantity | value |
|---|---|
| evr | 0.8858, 0.07317, 0.03278 |
| load[0] | -0.2184, -0.3338, -0.03369, -0.2851, 0.8709 |
| load[1] | 0.03458, -0.5574, 0.8013, -0.07872, -0.1997 |
| load[2] | -0.2197, -0.4488, -0.2104, 0.8393, 0.03956 |

## p2_pca_sens.json

Source: `Manuscripts/results/p2_pca_sens.json`; 10 values. Workbook sheet `R_p2_pca_sens`.

| quantity | value |
|---|---|
| sens.0.001 | 0.8858, 0.07317, 0.03278, 0.008299 |
| sens.0.1 | 0.7435, 0.1732, 0.06629, 0.01701 |
| sens.1 | 0.6274, 0.2481, 0.09837, 0.02614 |
| sens.5 | 0.565, 0.2511, 0.1481, 0.03581 |
| sens.10 | 0.5233, 0.271, 0.168, 0.03776 |
| clip1_evr | 0.6274, 0.2481, 0.09837 |
| clip1_load[0] | -0.1982, -0.3672, 0.06212, -0.338, 0.8413 |
| clip1_load[1] | 0.08674, -0.5565, 0.7747, -0.01804, -0.2869 |
| clip1_load[2] | -0.2647, -0.4327, -0.2259, 0.826, 0.09732 |
| cor_PC1 | 0.4828, 0.454, 0.9791 |

## p2_robust.json

Source: `Manuscripts/results/p2_robust.json`; 121 values. Workbook sheet `R_p2_robust`.

| quantity | value |
|---|---|
| terms | lnW, n, lnI1 |
| bootstrap.n_resamples | 2000 |
| bootstrap.law_loeo | 0.8982 |
| bootstrap.gbm3_loeo | 0.8662 |
| bootstrap.diff | 0.03198 |
| bootstrap.diff_mean | 0.03612 |
| bootstrap.ci95 | 0.004383, 0.0867 |
| without_E8_E9.n_runs | 8054 |
| without_E8_E9.n_experiments | 15 |
| without_E8_E9.loeo | 0.8818 |
| without_E8_E9.loso | 0.8811 |
| without_E8_E9.rmse_mm | 65.88 |
| without_E8_E9.mape_pct | 5.787 |
| without_E8_E9.steps[0].terms | lnW |
| without_E8_E9.steps[0].loeo | 0.5998 |
| without_E8_E9.steps[0].rmse_mm | 137.9 |
| without_E8_E9.steps[1].terms | lnW, n |
| without_E8_E9.steps[1].loeo | 0.8567 |
| without_E8_E9.steps[1].rmse_mm | 73.84 |
| without_E8_E9.steps[2].terms | lnW, n, lnI1 |
| without_E8_E9.steps[2].loeo | 0.8818 |
| without_E8_E9.steps[2].rmse_mm | 65.88 |
| without_E8_E9.gbm3_loeo | 0.7781 |
| table1.rain_mm.mean | 417 |
| table1.rain_mm.sd | 143.3 |
| table1.rain_mm.q25 | 297.4 |
| table1.rain_mm.median | 464.4 |
| table1.rain_mm.q75 | 464.4 |
| table1.irrigation_mm.mean | 179.7 |
| table1.irrigation_mm.sd | 191.4 |
| table1.irrigation_mm.q25 | 0 |
| table1.irrigation_mm.median | 178.2 |
| table1.irrigation_mm.q75 | 268.4 |
| table1.W.mean | 596.7 |
| table1.W.sd | 251 |
| table1.W.q25 | 464.4 |
| table1.W.median | 654.5 |
| table1.W.q75 | 722.1 |
| table1.ET_mm.mean | 546.1 |
| table1.ET_mm.sd | 195.7 |
| table1.ET_mm.q25 | 385.9 |
| table1.ET_mm.median | 611.5 |
| table1.ET_mm.q75 | 670.7 |

| quantity | value |
|---|---|
| table1.drainage_1m_mm.mean | 60.09 |
| table1.drainage_1m_mm.sd | 105.8 |
| table1.drainage_1m_mm.q25 | 2.803 |
| table1.drainage_1m_mm.median | 33.51 |
| table1.drainage_1m_mm.q75 | 77.62 |
| table1.R.mean | 163.7 |
| table1.R.sd | 96.47 |
| table1.R.q25 | 102.8 |
| table1.R.median | 129 |
| table1.R.q75 | 222.5 |
| R_by_experiment.E0_Baselines.runs | 10 |
| R_by_experiment.E0_Baselines.R_median | 106.9 |
| R_by_experiment.E0_Baselines.R_ET_median | 0.2398 |
| R_by_experiment.E0_Baselines.dS_W_median | -0.02058 |
| R_by_experiment.E10_Amount_Timing_Soil.runs | 954 |
| R_by_experiment.E10_Amount_Timing_Soil.R_median | 118.3 |
| R_by_experiment.E10_Amount_Timing_Soil.R_ET_median | 0.2295 |
| R_by_experiment.E10_Amount_Timing_Soil.dS_W_median | -0.01078 |
| R_by_experiment.E11_Temp_Evap_Rain.runs | 1000 |
| R_by_experiment.E11_Temp_Evap_Rain.R_median | 296.3 |
| R_by_experiment.E11_Temp_Evap_Rain.R_ET_median | 0.3751 |
| R_by_experiment.E11_Temp_Evap_Rain.dS_W_median | -0.009339 |
| R_by_experiment.E12_Sensor_Trigger_Rain.runs | 1000 |
| R_by_experiment.E12_Sensor_Trigger_Rain.R_median | 217.6 |
| R_by_experiment.E12_Sensor_Trigger_Rain.R_ET_median | 0.3186 |
| R_by_experiment.E12_Sensor_Trigger_Rain.dS_W_median | -0.008244 |
| R_by_experiment.E13_N_Water_Soil.runs | 975 |
| R_by_experiment.E13_N_Water_Soil.R_median | 124.5 |
| R_by_experiment.E13_N_Water_Soil.R_ET_median | 0.284 |
| R_by_experiment.E13_N_Water_Soil.dS_W_median | -0.006363 |
| R_by_experiment.E14_Mulch_Rain.runs | 1000 |
| R_by_experiment.E14_Mulch_Rain.R_median | 262.1 |
| R_by_experiment.E14_Mulch_Rain.R_ET_median | 0.409 |
| R_by_experiment.E14_Mulch_Rain.dS_W_median | 0.0008958 |
| R_by_experiment.E15_Method_Depth_Amount.runs | 1000 |
| R_by_experiment.E15_Method_Depth_Amount.R_median | 103.2 |
| R_by_experiment.E15_Method_Depth_Amount.R_ET_median | 0.1689 |
| R_by_experiment.E15_Method_Depth_Amount.dS_W_median | 0.003355 |
| R_by_experiment.E16_Edge_Cases.runs | 740 |
| R_by_experiment.E16_Edge_Cases.R_median | 157 |
| R_by_experiment.E16_Edge_Cases.R_ET_median | 0.2964 |
| R_by_experiment.E16_Edge_Cases.dS_W_median | -0.01008 |
| R_by_experiment.E1_Doses.runs | 189 |
| R_by_experiment.E1_Doses.R_median | 117.6 |
| R_by_experiment.E1_Doses.R_ET_median | 0.2123 |
| R_by_experiment.E1_Doses.dS_W_median | -0.001241 |
| R_by_experiment.E2_TriggerPressure.runs | 197 |
| R_by_experiment.E2_TriggerPressure.R_median | 167.5 |

| quantity | value |
|---|---|
| R_by_experiment.E2_TriggerPressure.R_ET_median | 0.2979 |
| R_by_experiment.E2_TriggerPressure.dS_W_median | -0.008657 |
| R_by_experiment.E3_SensorDepth.runs | 200 |
| R_by_experiment.E3_SensorDepth.R_median | 201.8 |
| R_by_experiment.E3_SensorDepth.R_ET_median | 0.3213 |
| R_by_experiment.E3_SensorDepth.dS_W_median | -0.008657 |
| R_by_experiment.E4_Methods.runs | 96 |
| R_by_experiment.E4_Methods.R_median | 103.4 |
| R_by_experiment.E4_Methods.R_ET_median | 0.1863 |
| R_by_experiment.E4_Methods.dS_W_median | -0.003341 |
| R_by_experiment.E5_SHP_Dynamics.runs | 300 |
| R_by_experiment.E5_SHP_Dynamics.R_median | 161.5 |
| R_by_experiment.E5_SHP_Dynamics.R_ET_median | 0.2882 |
| R_by_experiment.E5_SHP_Dynamics.dS_W_median | -0.007413 |
| R_by_experiment.E6_Mediterranean_Pollution.runs | 193 |
| R_by_experiment.E6_Mediterranean_Pollution.R_median | 70.43 |
| R_by_experiment.E6_Mediterranean_Pollution.R_ET_median | 0.1435 |
| R_by_experiment.E6_Mediterranean_Pollution.dS_W_median | -0.01191 |
| R_by_experiment.E7_Mulching.runs | 200 |
| R_by_experiment.E7_Mulching.R_median | 226.3 |
| R_by_experiment.E7_Mulching.R_ET_median | 0.4091 |
| R_by_experiment.E7_Mulching.dS_W_median | -0.005667 |
| R_by_experiment.E8_Rain_Intensity_Moisture.runs | 1000 |
| R_by_experiment.E8_Rain_Intensity_Moisture.R_median | 39.29 |
| R_by_experiment.E8_Rain_Intensity_Moisture.R_ET_median | 0.1438 |
| R_by_experiment.E8_Rain_Intensity_Moisture.dS_W_median | -0.1335 |
| R_by_experiment.E9_Soil_Moisture_Rain.runs | 987 |
| R_by_experiment.E9_Soil_Moisture_Rain.R_median | 81.23 |
| R_by_experiment.E9_Soil_Moisture_Rain.R_ET_median | 0.3736 |
| R_by_experiment.E9_Soil_Moisture_Rain.dS_W_median | -0.01399 |

## p2_s02_excl.json

Source: `Manuscripts/results/p2_s02_excl.json`; 27 values. Workbook sheet `R_p2_s02_excl`.

| quantity | value |
|---|---|
| terms | lnW, n, lnI1 |
| full.loeo | 0.8982 |
| full.loso | 0.9037 |
| full.coef.const | 2.503 |
| full.coef.lnW | 0.6458 |
| full.coef.n | -0.2992 |
| full.coef.lnI1 | 0.02932 |
| full.R_mean | 163.7 |
| full.R_share_of_ET | 0.2998 |
| without_S02.n_runs | 9651 |

| quantity | value |
|---|---|
| without_S02.loeo | 0.8972 |
| without_S02.loso | 0.9037 |
| without_S02.coef.const | 2.461 |
| without_S02.coef.lnW | 0.6519 |
| without_S02.coef.n | -0.2972 |
| without_S02.coef.lnI1 | 0.02947 |
| without_S02.R_mean | 163.1 |
| without_S02.R_share_of_ET | 0.2988 |
| without_S02_S03.n_runs | 9145 |
| without_S02_S03.loeo | 0.8931 |
| without_S02_S03.loso | 0.8976 |
| without_S02_S03.coef.const | 2.548 |
| without_S02_S03.coef.lnW | 0.6419 |
| without_S02_S03.coef.n | -0.3168 |
| without_S02_S03.coef.lnI1 | 0.03039 |
| without_S02_S03.R_mean | 165.4 |
| without_S02_S03.R_share_of_ET | 0.2958 |

# S4. Where every table and figure comes from

| Item | How it is made | Script(s) | Input and output |
|---|---|---|---|
| Table 1 | Descriptive table (definitions, design, parameters or tests); no computation | - | |
| Figure 1 | Figure drawn by the script from the input data | Manuscripts/reproducibility/make_a4.py; Manuscripts/reproducibility/analysis/p2_figs.py; Manuscripts/reproducibility/analysis/p2_fig1_new.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p2/fig01_pipeline_A4.pdf |
| Table 2 | Descriptive table (definitions, design, parameters or tests); no computation | - | |
| Table 3 | Table typed from the result file (27 of its numbers found there) | Manuscripts/reproducibility/p3_js/run_drip.js; Manuscripts/reproducibility/analysis/p3_lit_figs.py; Manuscripts/reproducibility/analysis/make_supplement_data.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/drip_out.json |
| Figure 2 | Figure drawn by the script from the input data | Manuscripts/reproducibility/make_a4.py; Manuscripts/reproducibility/analysis/p1_figs_d.py; Manuscripts/reproducibility/analysis/p1_fig1_new.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p1/fig01_framework_A4.pdf |
| Figure 3 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p1_figs_a.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p1/fig02_hydraulics.pdf |
| Table 4 | Table written by the script (tex/data_table.tex) | Manuscripts/reproducibility/analysis/data_overview.py | Data/Comprehensive_Summary/00_summary_all_simulations |
| Figure 4 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/data_overview.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/data/fig_data_overview.pdf |
| Table 5 | Table typed in the manuscript from the result files of the script | p2_robust.py; p2_robust.js | Data/daisy_ensemble.csv.gz (10 190 runs) |

| Item | How it is made | Script(s) | Input and output |
|---|---|---|---|
| Table 6 | Table typed from the result file (11 of its numbers found there) | Manuscripts/reproducibility/audit_numbers.py; Manuscripts/reproducibility/analysis/p3_figs.py; Manuscripts/reproducibility/p3_js/e1_heldout.js | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/e12_pred.json |
| Figure 5 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p2_figfail.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p2/figS_failed.pdf |
| Table 7 | Descriptive table (definitions, design, parameters or tests); no computation | - | |
| Figure 6 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p1_figs_a.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p1/fig03_memory_grid.pdf |
| Figure 7 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p1_figs_b.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p1/fig04_fates.pdf |
| Table 8 | Table typed from the result file (55 of its numbers found there) | Manuscripts/reproducibility/p3_js/run_drip.js; Manuscripts/reproducibility/analysis/p3_lit_figs.py; Manuscripts/reproducibility/analysis/make_supplement_data.py; p1_core.py; p1_core.js | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/drip_out.json |
| Figure 8 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p1_figs_b.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p1/fig05_predictors.pdf |
| Figure 9 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p1_figs_b.py; p1_rate_law.py; p1_rate_law.js | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p1/fig06_mechanism.pdf |
| Table 9 | Table typed from the result file (8 of its numbers found there) | Manuscripts/reproducibility/p3_js/run_drip.js; Manuscripts/reproducibility/analysis/p3_lit_figs.py; Manuscripts/reproducibility/analysis/make_supplement_data.py; p1_rate_law.py; p1_rate_law.js | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/drip_out.json |
| Figure 10 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p1_figs_c.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p1/fig08_replication.pdf |
| Figure 11 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p1_figs_d.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p1/fig10_robustness.pdf |
| Table 10 | Table typed from the result file (14 of its numbers found there) | p1_stats.py; p1_figs_d.py; p1_rank.js | Data/daisy_ensemble.csv.gz (10 190 runs); Results/07_Derived_Data_Tables/pareto_runs.csv |
| Figure 12 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p1_figs_c.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p1/fig07_modifiers.pdf |
| Table 11 | Table typed from the result file (8 of its numbers found there) | Manuscripts/reproducibility/analysis/p1_figs_c.py; p1_modifiers.js | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/p1_modifiers.json |
| Figure 13 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p1_figs_e.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p1/fig11_e11_surface.pdf |
| Table 12 | Table typed from the result file (7 of its numbers found there) | Manuscripts/reproducibility/p3_js/run_drip.js; Manuscripts/reproducibility/analysis/p3_lit_figs.py; Manuscripts/reproducibility/analysis/make_supplement_data.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/drip_out.json |
| Figure 14 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p2_figs.py; p2_a.py; p2_laws.py; p2_a.js; p2_laws.js | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p2/fig02_invariants.pdf |
| Table 13 | Table typed from the result file (34 of its numbers found there) | Manuscripts/reproducibility/p3_js/run_drip.js; Manuscripts/reproducibility/analysis/p3_lit_figs.py; Manuscripts/reproducibility/analysis/make_supplement_data.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/drip_out.json |
| Figure 15 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p2_figs.py; p2_R.py; p2_R.js | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p2/fig03_hidden_term.pdf |

| Item | How it is made | Script(s) | Input and output |
|---|---|---|---|
| Table 14 | Table typed from the result file (22 of its numbers found there) | Manuscripts/reproducibility/run_all.sh; Manuscripts/reproducibility/audit_numbers.py; Manuscripts/reproducibility/analysis/p1_core.py; p2_robust.py; p2_robust.js | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/p1_core.json |
| Figure 16 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p2_fig4.py; Manuscripts/reproducibility/analysis/p2_figs.py; p2_a.py; p2_pca.js; p2_pca_sens.js | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p2/fig04_manifold.pdf |
| Table 15 | Table typed in the manuscript from the result files of the script | p2_a.py; p2_figs.py; p2_fig4.py; p2_pca.js; p2_pca_sens.js | Data/daisy_ensemble.csv.gz (10 190 runs) |
| Figure 17 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p2_fig7.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p2/fig07_fates_by_soil.pdf |
| Figure 18 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p2_figs2.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p2/fig05_ET_law.pdf |
| Table 16 | Table typed in the manuscript from the result files of the script | (see the reproducibility appendix of the paper) | Data/daisy_ensemble.csv.gz (10 190 runs) |
| Table 17 | Table typed from the result file (18 of its numbers found there) | Manuscripts/reproducibility/run_all.sh; Manuscripts/reproducibility/make_e12.py; Manuscripts/reproducibility/audit_numbers.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/e12.json |
| Table 18 | Table typed in the manuscript from the result files of the script | (see the reproducibility appendix of the paper) | Data/daisy_ensemble.csv.gz (10 190 runs) |
| Table 19 | Table typed from the result file (12 of its numbers found there) | Manuscripts/reproducibility/p3_js/run_drip.js; Manuscripts/reproducibility/analysis/p3_lit_figs.py; Manuscripts/reproducibility/analysis/make_supplement_data.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/drip_out.json |
| Table 20 | Table typed from the result file (3 of its numbers found there) | Manuscripts/reproducibility/analysis/p4_reach.py; Manuscripts/reproducibility/analysis/make_supplement_data.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/p4_tableA1_reach.csv |
| Table 21 | Table typed from the result file (23 of its numbers found there) | Manuscripts/reproducibility/p3_js/run_drip.js; Manuscripts/reproducibility/analysis/p3_lit_figs.py; Manuscripts/reproducibility/analysis/make_supplement_data.py; p2_et.py; p2_dpf.py; p2_figs2.py; p2_nested.py; p2_et.js; p2_dpf.js; p2_etlaw.js; p2_nested.js; p2_s02_excl.py; p2_s02_excl.js | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/drip_out.json |
| Table 22 | Table typed in the manuscript from the result files of the script | (see the reproducibility appendix of the paper) | Data/daisy_ensemble.csv.gz (10 190 runs) |
| Figure 19 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p2_fig6.py; p2_laws2.py; p2_laws2.js | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p2/fig06_compensation.pdf |
| Table 23 | Table typed from the result file (19 of its numbers found there) | Manuscripts/reproducibility/run_all.sh; Manuscripts/reproducibility/analysis/p2_fig6.py; Manuscripts/reproducibility/analysis/p2_laws2.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/p2_laws2.json |
| Table 24 | Table typed from the result file (24 of its numbers found there) | Manuscripts/reproducibility/run_all.sh; Manuscripts/reproducibility/make_e12.py; Manuscripts/reproducibility/audit_numbers.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/e12.json |
| Figure 20 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p1_fig14_ensemble.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p1/fig14_ensemble_structure.pdf |
| Figure 21 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p1_fig13_time.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p1/fig13_time_management.pdf |
| Figure 22 | Figure drawn by the script from the input data | Manuscripts/reproducibility/run_all.sh; Manuscripts/reproducibility/analysis/p1_fig12_drivers.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p1/fig12_drivers.pdf |

| Item | How it is made | Script(s) | Input and output |
|---|---|---|---|
| Table 25 | Descriptive table (definitions, design, parameters or tests); no computation | - | |
| Table 26 | Table typed from the result file (8 of its numbers found there) | Manuscripts/reproducibility/audit_numbers.py; Manuscripts/reproducibility/analysis/p3_figs.py; Manuscripts/reproducibility/p3_js/e1_heldout.js | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/e12_pred.json |
| Figure 23 | Figure drawn by the script from the input data | Manuscripts/reproducibility/fig_vg_check.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p1/fig15_vg_check.pdf |
| Figure 24 | Screen capture of SWAFL Desktop 1.1.0 (not regenerated) | - | Manuscripts/figures/shots_jpg/real_evap_pore.jpg; Manuscripts/figures/shots_jpg/real_evap_flux_cells.jpg; Manuscripts/figures/shots_jpg/real_tillage_micro.jpg |
| Figure 25 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p1_figs_c.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p1/fig09_recoverable.pdf |
| Table 27 | Table typed in the manuscript from the result files of the script | (see the reproducibility appendix of the paper) | Data/daisy_ensemble.csv.gz (10 190 runs) |
| Table 28 | Table typed from the result file (34 of its numbers found there) | Manuscripts/reproducibility/run_all.sh; Manuscripts/reproducibility/audit_numbers.py; Manuscripts/reproducibility/analysis/p1_core.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/p1_core.json |

# References

Yechezkel Mualem. A new model for predicting the hydraulic conductivity of unsaturated porous media. *Water Resources Research*, 12:513–522, 1976. doi: 10.1029/WR012i003p00513.